

NEET Chapter Page – Thermodynamics

Introduction to Thermodynamics

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Thermodynamics is a branch of science that investigates heat, work, and temperature, as well as its connections to energy, radiation, and physical properties of the matter. The four laws of thermodynamics govern the behaviour of these values. It provides a quantitative description based on quantifiable macroscopic physical quantities, but statistical mechanics may be able to describe it in terms of microscopic elements.

Thermodynamics is the study of heat and energy with various physical transformations and chemical processes. It deals with the study of the energy transfer of a system with its surroundings. Energy in the form of heat can be absorbed or released by a reaction, and phase transitions like boiling and melting can do the same. The entropy of a reaction is used to assess whether it is favourable or unfavourable, spontaneous or nonspontaneous. It can also be used to calculate how much of each product and reactant is required in a complete reaction.

Thermodynamics is a crucial component of the JEE exam. The themes of an open system, closed system, isolated system, thermodynamic rules, thermodynamics formulae and so on are all important topics for the JEE exam. These topics are discussed in this article. This article is also used as thermodynamics chemistry notes for [NEET](#) aspirants. This article illustrates the types of questions that might be asked from this topic along with important thermodynamics notes.

Important Topics of Thermodynamics

- Thermodynamic Processes.
- First Law of Thermodynamics.
- Second Law of Thermodynamics.
- Third Law of Thermodynamics.
- Enthalpy and Enthalpy Change.
- Specific and Molar Heat Capacity.
- Carnot Cycle.
- Free Energy.
- Exothermic Reaction.
- Endothermic Reaction.
- Heat of Reaction.

Important Definitions of Thermodynamics

Important Topics	Explanation
System, surroundings and boundary	The surroundings refer to the remainder of the universe that is not the part of the system whereas the system refers to the piece of the universe that is being seen. This is how a thermodynamic system is defined. The imaginary line which separates the system with its surroundings is called boundary.
Isolated system	There is no interaction between the system and its surroundings in this case.
Closed system	This type of system can interchange energy with its surroundings in the form of heat, work, or radiation, but no matter what it will get exchanged.
Open system	This type of system can interact with its surroundings in terms of both matter and energy.
Intensive properties	Intensive properties are those properties that are unaffected by the amount of matter in the system or the system's size.
Extensive properties	The extent of extensive properties is proportional to the amount of matter present in the system.

State of a System and State Variables

- Macroscopic qualities that specify the state of a system include state variables, state functions, and thermodynamic parameters.
- The state qualities are only influenced by the system's starting and ending states.
- It is unaffected by the method used to implement the change.

Thermodynamic Equilibrium

- A system is said to have reached thermodynamic equilibrium when it displays no additional propensity to change its property over time.
- The thermodynamic equilibrium criterion specifies that all the three equilibrium forms must occur simultaneously in a system.

Chemical Equilibrium: A system in which the system's composition is continuous and definite.

Mechanical Equilibrium: There is no chemical interaction between the system's components or between the system and its surroundings.

Maintain constant pressure in order to succeed.

Thermal Equilibrium: It is conceivable if the temperature remains constant, that is if there's no heat transfer between the system and the surroundings.

Thermodynamic Processes

- The transition of a thermodynamic system from one state to another is referred to as a thermodynamic process. This process covers a number of processes.

Isothermal Process

- In this procedure, the process is carried out at a constant temperature. $E = 0$ follows from $dT = 0$.

Adiabatic Process

- In this method, there is no heat exchange between the system and its surroundings.
- With $dQ = 0$ and insulated borders, the system is thermally isolated.

Isobaric Process

- Throughout the shift in the process, the pressure remains constant, i.e., $dP = 0$.
- For example, when water boils, it produces steam, and when it freezes, it produces ice.

Isochoric Process

- Throughout the shift in the process, the volume remains constant, i.e., $dV = 0$.
- For example, when a gasoline-air mixture is burned in a car engine, the temperature and pressure of the gas inside the engine rise. Meanwhile, the gas volume remains constant.

Cyclic Process

- When a system goes through a succession of procedures before returning to its original state, it is called a cyclic process.
- For a cyclic process, $dE = 0$ and $dH = 0$.

Reversible Process

- Reversible process in thermodynamics is the process that is reversible in nature. But, while reversing the process there should not any changes made to the environment by system or its surroundings.
- For example, springs that can be extended and get back to their original position. Electrolysis is the frictionless mobility of solids (with no resistance in the electrolyte).

Irreversible Process

- When a process goes from its beginning condition to its final state in a single step in a fixed period of time and cannot be reversed, it is called an irreversible process.
- The quantity of entropy grows in an irreversible process.
- There are many irreversible processes in nature.
- All the natural processes are irreversible in nature.
- Examples of irreversible processes include the flow of electric current through a conductor with resistance.

Internal Energy

- Every system has some amount of matter which possesses a definite amount of energy, hence the energy contained within the system is the energy contained within the system.
- This energy is a state function that represents the sum of all energy forms (chemical, mechanical, electrical, etc.).
- It is represented by the letter/symbol E .

First Law of Thermodynamics

- It is also known as the law of energy conservation.
- It states, *"Energy can neither be created nor destroyed although it can be converted from one form into another."*
- Change in internal energy = Heat added to the system + Work done on the system.

$$E_2 - E_1 = \Delta E = q + w$$
- It was conceived by Helmholtz and Robert Mayer.
- The internal energy of a system is depleted when it does work (w) on the environment.
- Change in internal energy = Heat added to the system – Work done by the system

$$\Delta E = q + (-w) = q - w$$

Enthalpy and Enthalpy Change

- The heat content of a system under constant pressure is called enthalpy.
- The letter 'H' stands for Enthalpy.
- From the first law of thermodynamics,
 $q = E + W$; $q = E + P\Delta V$
 It is possible to provide a heat change at a constant pressure,

$$q = E + P\Delta V.$$

Heat may be replaced at enthalpy at constant pressure.

$$\Delta H = E + P\Delta V.$$

The heat of reaction (in a chemical process) or the heat change under constant pressure is denoted by the letter H.

Specific Heat Capacity and Molar Heat Capacity

- The amount of heat required to raise the temperature of one gram of a substance by one degree celsius is known as its specific heat capacity.

$$q = mc\Delta T; (c_p = \text{constant pressure}/c_v = \text{constant volume})$$

- The amount of heat required to raise the temperature of one mole of a substance by one degree celsius is known as its molar heat capacity.
- Gases have a tendency to expand when heated under constant pressure, necessitating an additional energy source to raise their temperature by 1°C when necessary under constant volume conditions.
- Therefore, $C_p > C_v$.
- $C_p = C_v + \text{Work done in expansion}$; Work done in expansion is $P\Delta V (=R)$; C_v = molar heat capacities at constant volume, C_p = molar heat capacities at constant pressure.
- Relations between C_p and C_v .

Parameter	Mathematical Relation
General Relation	$C_p - C_v = R = 2 \text{ calories} = 8.314 \text{ J.}$
For monatomic gases	$C_v = 3/2R$
For diatomic gases	$C_v = 3/2 + x$; x = variable from gas to gas.
Ratio of Molar Capacities, (γ)	C_p/C_v
Ratio of Molar Capacities, For monatomic gases, (γ)	$C_p/C_v = \{(5/2)R\}/\{(3/2)R\} = 1.66$
Ratio of Molar Capacities, For diatomic gases, (γ)	$C_p/C_v = \{(7/2)R\}/\{(5/2)R\} = 1.40$
Ratio of Molar Capacities, For triatomic gases, (γ)	$C_p/C_v = 8R/6R = 1.33$

Second Law of Thermodynamics

- All of the limits imposed by the first law are removed by the second law of thermodynamics.
- This law shows that spontaneous processes have a positive overall energy change.

The Carnot Cycle

- Carnot cycle is a type of cyclic process in which the net work done equals the heat absorbed.
- Thus, the bigger the temperature differential between the high and low temperature reservoirs, the more heat is generated, which is converted into work by the heat engine.
- Thermodynamic efficiency = $w/q_2 = (T_2 - T_1)/T_1$.

Entropy

- Entropy is a metric that measures the unpredictability or disorder of a system's molecules.
- It's a metric for assessing the state of the thermodynamic system.
- $S_{\text{final}} - S_{\text{initial}} = \Delta S = q_{\text{rev}}/T$.
- If heat is absorbed, then $S = +ve$ and if heat is evolved, then, $\Delta S = -ve$.

Free Energy

- Gibbs free energy (G) is a unit of measurement for the maximum work done by a reversible process under constant pressure and temperature, also known as useful work done.
- By subtracting the system's enthalpy from the product of absolute temperature and entropy, we can find Gibbs free energy, i.e., $\Delta G = \Delta H - T\Delta S$.

Third Law of Thermodynamics

- This law states, "The entropy of all perfectly crystalline solids approaches zero at the temperature approaches absolute zero."
- Because entropy is a measure of disorder, a perfectly crystalline material has a perfect order of its constituent particles at absolute zero.

Exothermic and Endothermic Reactions

- Exothermic reactions are those that occur as a result of heat energy evolution.

$$\Delta H = H_p - H_r; H_p < H_r; \Delta H = -ve.$$

- Examples of Exothermic Reactions

Example	Equations & Parameter Values
Fermentation	$- C_6H_{12}O_6 \text{ (glucose)} \rightarrow 2 C_2H_5OH \text{ (ethanol)} + 2 CO_2 \text{ (carbon dioxide)}$ $- C_6H_{12}O_6 \text{ (glucose)} \rightarrow 2 CH_3CHOHCOOH \text{ (lactic acid)}$
Formation of CO_2 gas	$C(s) + O_2(g) \rightarrow CO_2(g) + 393.5 \text{ kJ}$ $C(s) + O_2(g) \rightarrow CO_2(g); \Delta H = - 393.5 \text{ kJ}$

- Endothermic reactions are those that take place when heat energy is absorbed.

$$\Delta H = H_p - H_r; H_p > H_r; \Delta H = +ve.$$

- Examples of Endothermic Reactions

Example	Equation & Parameter Values
Formation of NO	$N_2 + O_2 \rightarrow 2NO; \Delta H = + 180.5 \text{ kJ}$

Heat of a Reaction or Enthalpy of a Reaction

- The heat of a reaction, also known as reaction enthalpy, is the difference between the enthalpies of the products and reactants acquired after the reactants reacted entirely in a chemical process.
- Enthalpy of reaction (heat of reaction) = $\Delta H = \Sigma H_p - \Sigma H_r$.

Solved Examples from the Chapter

Example 1: The entropy of a system approaches a constant value as the temperature _____.

- a. Increases
- b. Decreases
- c. Approaches absolute zero
- d. None of the options

Solution:

- When the temperature approaches absolute zero, the system approaches the constant value, according to the third rule of thermodynamics.
- Hence, the correct answer is option (c) Approaches absolute zero.

Key point to remember: Third Law of Thermodynamics: The entropy of all perfectly crystalline solids approaches zero when the temperature approaches absolute zero

Example 2: Which of the following statements is correct?

- a. An open system is one in which reacting species are present in a covered beaker.
- b. In a closed system, there is an exchange of energy and matter between the system and its surroundings.
- c. The presence of reactants in a closed vessel made up of copper is an example of a closed system.
- d. A closed system is defined as the presence of reactants in a thermos flask or any other closed insulated vessel.

Solution:

- In a closed system, there is no interchange of substances (for example, the existence of reactants in a closed vessel made of a conducting material like copper).
- But there is an exchange of energy between the system and its surroundings.
- Hence, the correct answer is (c). The presence of reactants in a closed vessel made up of copper is an example of a closed system.

Key point to remember: Definition of a closed system. In a closed type of system, there can be an interchange of energy with its surroundings in the form of heat, work, or radiation, but not matter.

Solved Examples from Previous Year Questions

Question 1: The internal energy change in a system that has absorbed 2 kJ of heat and done 500 J of work is:

- a. 6400 J
- b. 5400 J
- c. 7900 J
- d. 8900 J

Solution:

- According to the first law of thermodynamics,
$$Q = \Delta U + W,$$

$$\begin{aligned}\therefore \Delta U &= Q - W \\ &= 2 \times 4.2 \times 1000 - 500 \\ &= 8400 - 500 \\ &= 7900 \text{ J}\end{aligned}$$

- Hence, the correct answer is option (c) 7900 J

Trick: First Law of thermodynamics: Change in internal energy = Heat added to the system + Work done on the system.

Question 2: A gas is compressed isothermally to half its initial volume. The same gas is compressed separately through an adiabatic process until its volume is again reduced to half. Then :

- Compressing the gas isothermally will require more work to be done.
- Compressing the gas through an adiabatic process will require more work to be done.
- Compressing the gas isothermally or adiabatically will require the same amount of work.
- Which of the case (whether compression through isothermal or through adiabatic process) requires more work will depend upon the atomicity of the gas

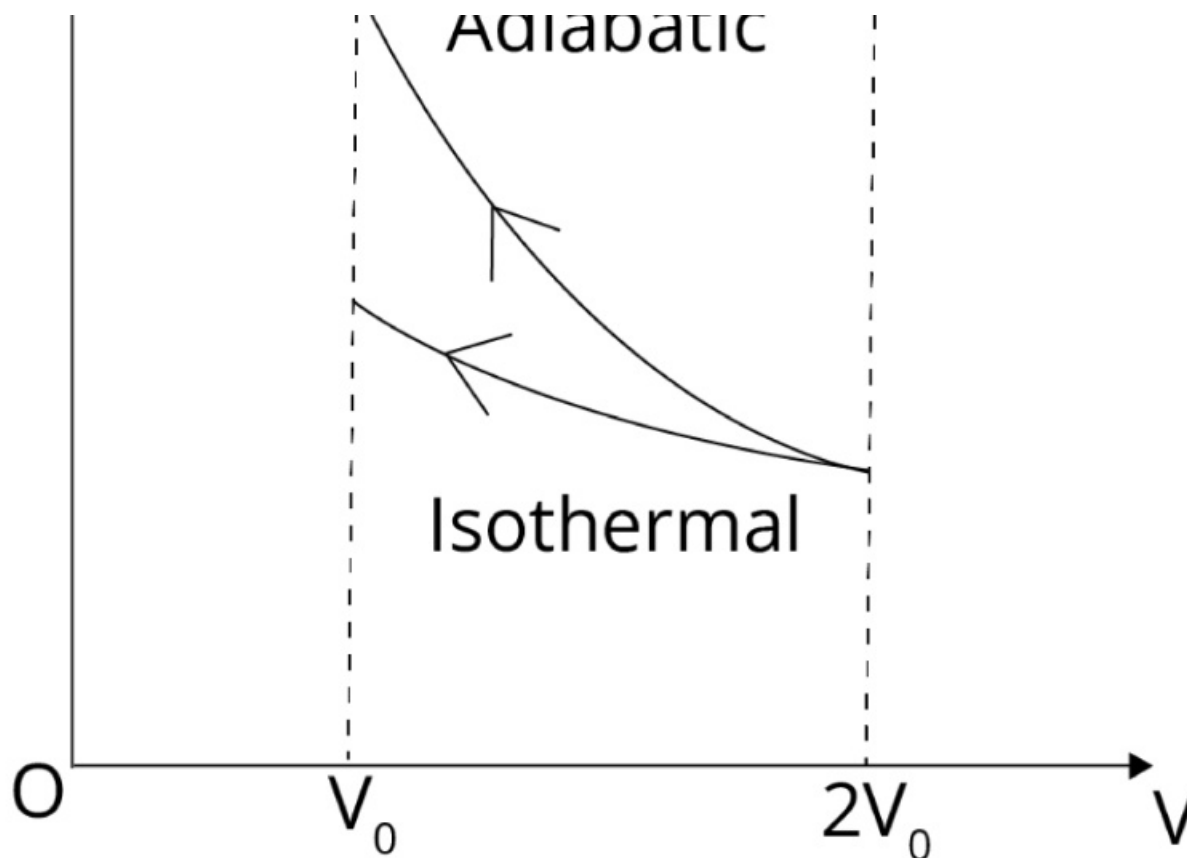
Solution:

- The external work done is the negative of the area with the volume axis from the figure given below:



P

Adiabatic



- Also, the adiabatic work done is greater than the isothermal work done.
Work Done (adiabatic) > Work Done (isothermal).
- Hence, the correct answer is (b). Compressing the gas through an adiabatic process will require more work to be done.

Trick: The isothermal process is carried out at a steady temperature in this procedure. $E = 0$ is the result of $dT = 0$. There is no heat exchange between the system and its surroundings in an adiabatic process. The adiabatic system is thermally isolated with $dQ = 0$ and insulated borders.

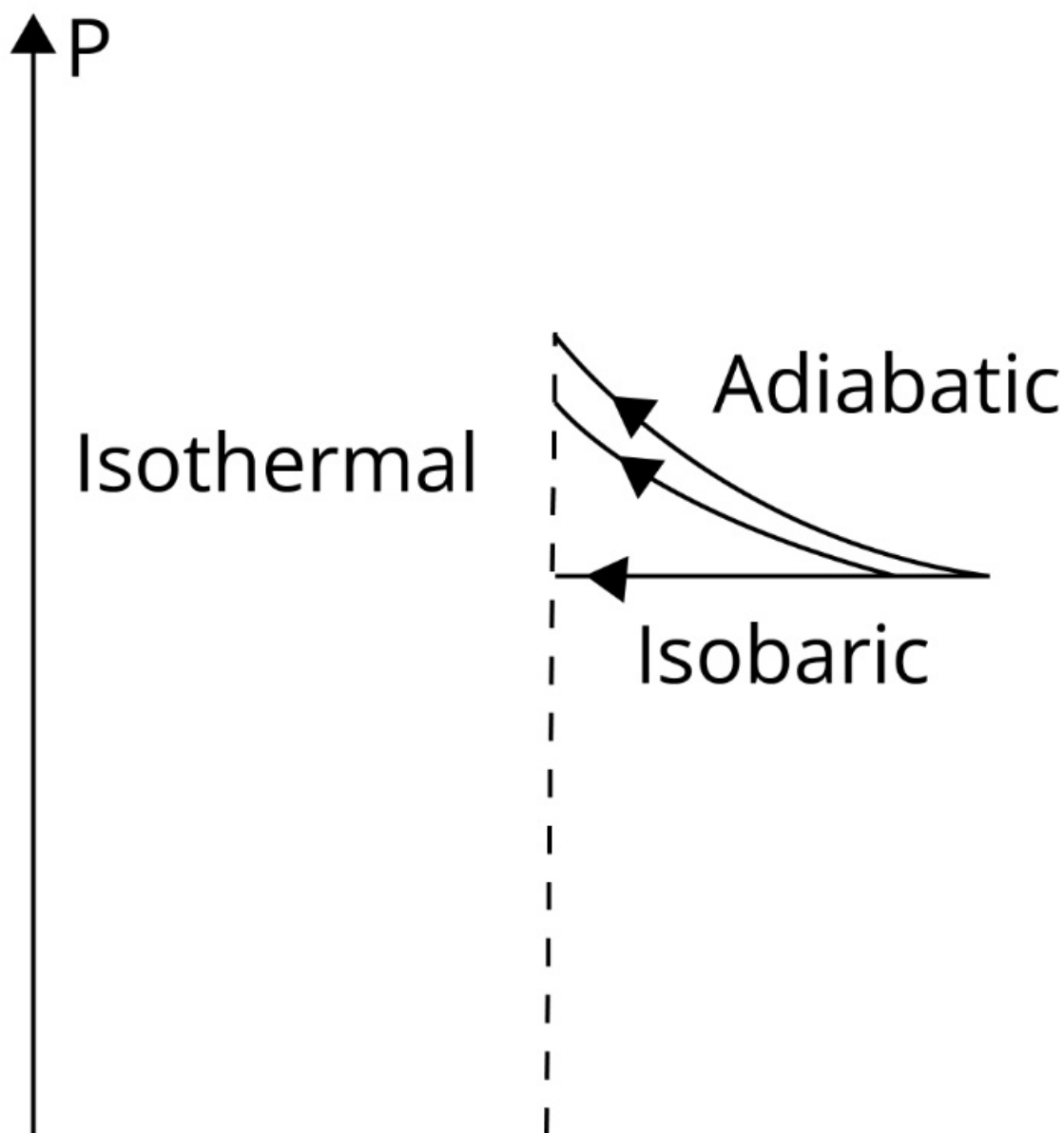
Question 3: An ideal gas is compressed to half its initial volume by means of

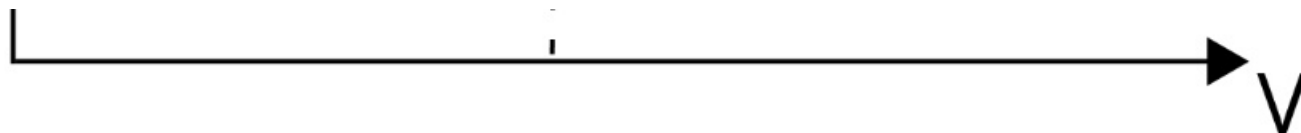
several processes. Which of the processes results in the maximum work done on the gas?

- a. Isobaric
- b. Isochoric
- c. Isothermal
- d. Adiabatic

Solution:

- Because the area under the curve is the greatest for an adiabatic process, the work done on the gas ($W = PdV$) is greatest for an adiabatic process.





- Hence, the correct answer is (d) Adiabatic.

Trick: There is no heat exchange between the system and its surroundings in the adiabatic process. The system is thermally isolated with $dQ = 0$ and insulated borders.

Practice Questions

Question 1: The entropy of a flawless crystal is 0 at absolute zero. In thermodynamics it is referred to as:

- The First Law
- The Second Law
- The Third Law
- None of the Above.

Answer: (c) The Third Law.

Question 2: The direction of any chemical reaction at constant pressure and temperature is one in which the _____ decreases.

- Entropy
- Gibbs free energy
- Enthalpy
- None of the above

Answer: (b) Gibbs free energy.

Conclusion

Thermodynamics is the study of how one body interacts with another in terms of heat and work. In this context, a system is defined as a quantity of substance or a location in space under investigation. Anything that exists outside of the system is referred to as the environment.

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